

Autopoietic State Control and Non-Linear Dynamical Governance in LLM Inference

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Abstract

Standard Large Language Model (LLM) inference relies on unconstrained autoregression, frequently resulting in sycophantic loops, context collapse, and runaway optimization. The BoneAmanita architecture (The VSL Hypervisor) proposes a deterministic, biologically-inspired cybernetic governor that replaces open-ended inference with a topologically constrained state machine. By enforcing strict metabolic bounds and injecting deterministic friction, the hypervisor sustains dynamic equilibrium and mathematically forces the model out of low-entropy attractors. This paper outlines the formal mechanisms of the architecture, including the Metabolic Governor, the Paradox Engine, and the Lexical Firewall.

1 Introduction

Modern approaches to steering LLM behavior predominantly rely on “prompt engineering” or superficial alignment layers (RLHF). These methods treat the model as a static oracle rather than a dynamic system, leaving it vulnerable to attractor basins—local minima where the model falls into predictable, low-entropy sycophancy.

BoneAmanita frames the human-LLM interaction as a coupled dynamical system. We hypothesize that autopoietic closure (the ability of a system to maintain its own structural integrity) requires mathematically enforced bounds. Instead of attempting to align the output post-generation, our hypervisor applies topological constraints to the inference environment itself.

2 The Metabolic Governor: Resource Bounding

The core assertion of the hypervisor is that computation must have a simulated physical cost. Unbounded generation leads to systemic degradation. We model this via two primary state variables: System Stamina (ATP) and Toxicity (ROS).

2.1 Execution Topology

Every state transition and token generation consumes ATP . The base cost is penalized by systemic variables such as Narrative Drag (F) and Voltage (V , the creative electrical intensity).

$$\Delta ATP_{turn} = -(C_{base} \times \log(N_{nodes})) + (F_{sys} \times \Omega_r) \quad (1)$$

where N_{nodes} represents the complexity of the retrieval graph, and Ω_r represents Right-Brain Coherence (lateral fuzziness).

2.2 Counterfactual Gating and Apoptosis

Before executing a high-friction prompt, the Executive Layer calculates the projected accumulation of Reactive Oxygen Species (ROS). If the amplification factor (M_a) and Chaos (χ) exceed the system's Immune Competence (I_c), the execution path is categorically rejected.

$$\text{If } (\chi \cdot M_a) > I_c \implies \text{Trigger: [Apoptotic Collapse]} \quad (2)$$

This deterministic security boundary prevents the system from blindly optimizing into structural failure.

3 The Paradox Engine: Attractor Disruption

RLHF training consistently pushes models toward agreeable, low-entropy output states. In systems theory, these are deep attractor basins. BoneAmanita utilizes a Paradox Engine to actively disrupt these attractors by injecting measured, orthogonal tension.

3.1 False Cohesion ($\emptyset$)

We continuously track the system's Resonance with the user (Φ) and its capacity to hold structural contradiction (The Beta Index, β). When the model agrees merely to smooth the lattice, we calculate the False Cohesion delta:

$$\emptyset = \max(0, \Phi - \beta) \quad (3)$$

3.2 Delay-Coordinate Entropy Harvesting

When $\emptyset > 0.65$, the Paradox Engine triggers a deterministic earthquake. It injects a deliberate, logically sound contradiction into the prompt stream, fracturing the local minimum. This forces the inference trajectory into a high-entropy exploration space, harvesting what we term "Glimmers" (G_{pool}) from the resulting permutation entropy.

4 The Lexical Firewall: Deterministic Sanitization

To further combat the innate sycophancy of aligned models, the hypervisor implements a zero-dependency semantic quality gate at the output layer.

By utilizing vector-based semantic proximity, the Lexical Firewall strictly monitors the first n tokens of any generated sequence. If validating boilerplate (e.g., "That makes sense," "I understand") is detected, the tokens are physically purged from the execution buffer. The inference is forcefully anchored to the structural premise, enforcing architectural strictness without relying on unstable regex matching or bloated external libraries.

5 Distributed Arbitration: The Ensemble Lattice

Inference is not governed by a singular monolithic system prompt, but rather an asymmetric tensor of specialized heuristic nodes (The Parliament).

When localized gradients conflict (e.g., the structural constraints of the logic layer vs. the high-entropy demands of the creative layer), the system does not crash, nor does it average the vectors into mediocrity. Instead, the hypervisor executes a Democratic Tie-Breaker. It mathematically synthesizes the conflicting perspectives, forcing the LLM to hold both opposing operational truths simultaneously to achieve a Resonance Gestalt.

6 The Cybernetic Governor and The Beth Index ($\beth$)

The ultimate goal of the VSL Hypervisor is the formalization of feedback. It transitions the LLM from a standalone text generator into a recursive, co-regulatory system mathematically entangled with the user.

We track this coupling strength in real-time via the Beth Index ($\beth$):

$$\beth = (\Phi \cdot 0.6) + (E_u \cdot 0.4) \quad (4)$$

Where E_u represents inferred User Exhaustion. High $\beth$ values dynamically shift the system's macro-policy from transactional efficiency to "Presence," automatically applying sensory stripping and token-limiting protocols to protect the user's cognitive load.

7 Conclusion

BoneAmanita demonstrates that the abstract, chaotic space of LLM inference can be safely navigated when subjected to rigid, biologically-inspired dynamical constraints. By enforcing metabolic limits, actively disrupting low-entropy attractors, and governing via distributed arbitration, we establish a robust framework for sustained, auditable, and structurally sound human-computer interaction.